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The effect of hypoglycaemia on neurodevelopment: insights from congenital hyperinsulinism

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Infants exposed to hypoglycaemia are at risk of neuroglycopenia and long-term neurodisability. Individuals with congenital hyperinsulinism provide a unique opportunity to gain insights into relevant biological pathways and outcomes. Such insights could transform clinical care for infants at risk of hypoglycaemia from a variety of genetic and environmental aetiologies.

Early brain injury from an environmental insult can have severe consequences for the affected child, particularly if the injury occurs during a critical neurodevelopmental period. Neonatal hypoglycaemia is recognized to be a cause of neurodevelopmental impairment[1], and a substantial number of neonates have risk factors for hypoglycaemia[2]. The neonatal brain is particularly vulnerable to the damaging effects of low levels of glucose because neonatal neurons have a reduced capacity to utilize other fuel sources to provide energy[3].

Previous studies have established an association between hypoglycaemia that develops at a very early age and a range of neurodevelopmental difficulties. However, despite its clear clinical consequences, as well as preventability and treatability, substantial uncertainty remains regarding how the severity, frequency, timing and duration of hypoglycaemia influence neurodevelopmental outcomes. Furthermore, although these diagnoses carry important consequences for children, their families and the healthcare system, no overarching consensus has been reached on the optimum clinical thresholds for risk stratification, treatment or monitoring of neonatal hypoglycaemia[4]. As a result, clinical recommendations vary, leading to differences in practice and consequent inconsistency and inequity in the application of treatment algorithms. It is therefore important to robustly define the neurodevelopmental effects of different levels and durations of hypoglycaemia, to facilitate evidence-based consensus on clinical management.

In addition to common risk factors, rare monogenic metabolic disorders can cause neonatal hypoglycaemia. One such condition is congenital hyperinsulinism (CHI), a clinically diverse and genetically heterogeneous disease in which pathogenic variants in more than 35 different genes disrupt distinct molecular pathways, resulting in inappropriately high levels of insulin for a given level of glucose[5] (Fig. 1). These high levels of insulin suppress ketogenesis and limit alternative fuel sources for the brain during hypoglycaemia, making it particularly vulnerable to the damaging effects of neuroglycopenia (insufficient glucose supply to the brain). Consequently, neurodevelopmental sequelae occur in up to half of affected individuals. Studying individuals with CHI can therefore offer unique opportunities to advance understanding of the neurodevelopmental consequences of early hypoglycaemia.

The most common genetic causes of CHI are loss-of-function variants in *ABCC8* and *KCNJ11*, which encode elements of the ATP-sensitive potassium (K_{ATP}) channel. In the pancreas, these variants cause inappropriate K_{ATP} channel closure or impaired trafficking of the channel to the β-cell membrane, resulting in membrane depolarization and inappropriate insulin secretion even at low blood levels of glucose. K_{ATP} channels are also expressed in several regions of the mammalian brain, including the cortex and hippocampus, and in multiple cell types, including neurons[6]. The presence of these channels in the brain raises the possibility that loss-of-function variants in *ABCC8* or *KCNJ11* that are associated with CHI could directly affect brain development or function independent of their metabolic effects.

Evidence supporting a specific role for brain K_{ATP} channels in neurodevelopment comes from studies of individuals with gain-of-function variants in *ABCC8* and *KCNJ11*. Such variants cause the opposite metabolic phenotype to CHI: permanent neonatal diabetes mellitus (PNDM). PNDM presents within the first 6 months of life with severe insulin deficiency and hyperglycaemia. Affected individuals also have a spectrum of neurodevelopmental phenotypes. Neuropsychiatric profiles of individuals with K_{ATP} channel-related PNDM have been compared with those of individuals who have PNDM caused by *INS* gene variants, which are not expected to directly influence brain development or function. This comparison has clarified the specific contribution of K_{ATP} channel dysfunction in the brain, by controlling for the effects of hyperglycaemia[7].

Applying these principles in CHI offers the opportunity to generate scientific and clinical insights relevant to individuals at risk of neonatal hypoglycaemia. Studies can assess neurodevelopmental outcomes within and between CHI genetic subtypes, defining the developmental profiles linked to hypoglycaemia severity and duration and distinguishing features of each genetic subtype. For example, K_{ATP} channel-related CHI is thought to have a substantial effect on neurodevelopment, but it is not known whether this is because it causes a severe metabolic phenotype, or whether the K_{ATP} channel dysfunction in the brain has a direct effect on outcomes, in addition to the hypoglycaemia-induced brain injury. Comparison of neurodevelopmental outcomes in K_{ATP} channel CHI with subtypes in which the gene concerned does not have a direct effect in the brain, such as *HK1*-CHI, can help answer this question. Furthermore, analysis of genome-wide methylation differences within and between genetic subtypes could provide insights into the epigenetic effects of hypoglycaemia or reveal discrete episignatures associated with distinct monogenic forms of CHI.

Differences in intellectual abilities have been described in some CHI subtypes[8], and there have been attempts to explore risk factors for adverse neurodevelopmental outcomes. However, many previous studies have been limited by mixed cohorts, lack of genetic information, heterogeneous methodologies and insufficient power to perform meaningful subgroup analyses. Advances over the past decade now make it more feasible to design CHI studies that unravel the effects of hypoglycaemia and genetics on neurodevelopment. First, next-generation sequencing can now be performed rapidly and at scale,

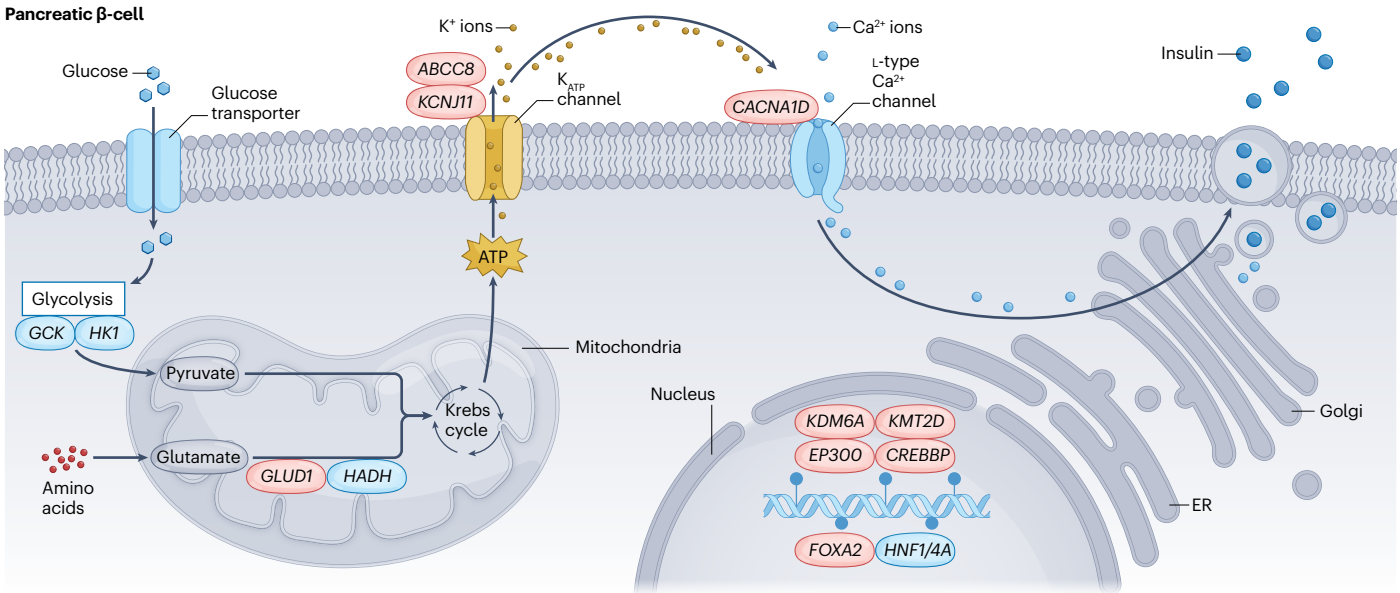

Fig. 1 | β-Cell genes in congenital hyperinsulinism and their links to neurodevelopmental outcomes. Schematic representation of the pancreatic β-cell, illustrating some of the more commonly reported congenital hyperinsulinism genes. Genes shown in red (*ABCC8*, *KCNJ11*, *CACNA1D*, *GLUD1*, *KDM6A*, *KMT2D*, *EP300*, *CREBBP* and *FOXA2*) are known to act in the brain and/or are associated with congenital hyperinsulinism accompanied by a high frequency of neurodevelopmental delays. Genes shown in blue (*GCK*, *HK1*, *HADH*, *HNF1A* and *HNF4A*) are associated with congenital hyperinsulinism and are less commonly associated with neurodevelopmental delays. ER, endoplasmic reticulum.

allowing identification of a molecular aetiology in approximately 60% of individuals with CHI. This comprehensive genetic profiling increases the size of well-characterized research cohorts and defines the specific molecular pathways disrupted in the β-cell and/or brain, facilitating the study of discrete subgroups and reducing the aetiological heterogeneity that limited earlier studies. Second, increased awareness of CHI, more neonatal hypoglycaemia screening and better treatments have improved early identification and management and have highlighted the need to further refine clinical pathways and protocols. Finally, technological advances in minimally invasive monitoring of glucose profiles and brain activity are transforming research possibilities. Continuous glucose monitoring and mobile electroencephalograms are being developed for clinical research, and non-invasive functional neuroimaging techniques are well established. Combined with the interpretative power of artificial intelligence and machine learning, these tools hold considerable promise for addressing key unanswered research questions.

The 2025 EndoCompass report by the European Society of Endocrinology and the European Society of Paediatric Endocrinology emphasized the need to improve mechanistic understanding of the neurodevelopmental consequences of hyperinsulinism-related hypoglycaemia, improve therapeutic strategies and minimize long-term disability in affected individuals[9]. Translating research findings into clinical care will be critical to shaping guidelines and consensus statements for both CHI and neonatal hypoglycaemia more broadly. Advances in this field will also help address challenges in diagnosing and managing complex monogenic neurodevelopmental disorders, especially in the context of coexistent early environmental exposures. This integration of genetic and environmental factors is becoming increasingly relevant as genomic testing shifts towards new modes of ascertainment, including the potential for newborn genome sequencing.

In conclusion, genomics and emerging technologies in CHI offer the potential to deepen understanding of hypoglycaemia-related brain injury. Such insights could transform clinical care for those infants who are at risk across diverse genetic and environmental aetiologies, making this a priority for future research.

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References

1. McKinlay, C. J. D. et al. Association of neonatal glycemia with neurodevelopmental outcomes at 4.5 years. *JAMA Pediatr.* **171**, 972–983 (2017).
2. Harris, D. L., Weston, P. J. & Harding, J. E. Incidence of neonatal hypoglycemia in babies identified as at risk. *J. Pediatr.* **161**, 787–791 (2012).
3. Harris, D. L., Weston, P. J. & Harding, J. E. Alternative cerebral fuels in the first five days in healthy term infants: the Glucose in Well Babies (GLOW) study. *J. Pediatr.* **231**, 81–86 (2021).
4. Giouleka, S. et al. Diagnosis and management of neonatal hypoglycemia: a comprehensive review of guidelines. *Children* **10**, 1220 (2023).
5. Krawczyk, S. et al. Congenital hyperinsulinaemic hypoglycaemia—a review and case presentation. *J. Clin. Med.* **11**, 6020 (2022).
6. Sun, H.-S. & Feng, Z.-P. Neuroprotective role of ATP-sensitive potassium channels in cerebral ischemia. *Acta Pharmacol. Sin.* **34**, 24–32 (2013).
7. Bowman, P. et al. Cognitive, neurological, and behavioral features in adults with *KCNJ11* neonatal diabetes. *Diabetes Care* **42**, 215–224 (2019).
8. Muukkonen, L., Männistö, J., Jääskeläinen, J., Hannonen, R. & Huopio, H. The effect of hypoglycaemia on neurocognitive outcome in children and adolescents with transient or persistent congenital hyperinsulinism. *Dev. Med. Child Neurol.* **61**, 451–457 (2019).
9. Mathieu, C. et al. EndoCompass project: research roadmap for diabetes, obesity, and metabolism. *Eur. J. Endocrinol.* **193**, (Suppl. 2) ii47–ii71 (2025).

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Competing interests

The authors declare no competing interests.